

ACTIVITY

1. Medical Image Enhancement
→ which gray-level transformation would you recommend?

A. Log Transformation.

2) why is it suitable?

Log transformation expands the range of dark pixel values and compresses the bright pixel values.

This makes darker tissues more visible without significantly affecting the brighter bone regions.

3) What effect will it have on the image?

The darker tissues become brighter and clearer, while the bones remain visible. Overall image contrast in dark regions is improved.

2. Night-Time Surveillance.

→ which transformation would improve the visibility?

Log Transformation.

2) why is it more suitable than negative transformation?

Since most pixel values are very low (0-40), log transformation enhances these dark intensity values and reveals hidden details.

Negative transformation only inverts brightness and does not effectively improve visibility in dark regions.

3) predict the appearance of the enhanced image.

The parking lot will appear brighter, with vehicles, people and surrounding objects becoming more

visible while maintaining natural brightness levels.

3. Fingerprint Recognition

→ which gray-level transformation would you apply?

Contrast stretching

2. explain why

Contrast stretching increases the difference b/w the fingerprint ridges and valleys, making the fine ridge patterns clearer and improving recognition accuracy.

3. what would happen if a log transformation were applied instead?

Log transformation would brighten the darker regions but would not significantly improve the contrast b/w ridges and valleys. As a result, the fingerprint details may remain less distinct compared to contrast stretching.